

Grammatical Features

based largely on Goddard (2005) Chapter 4.

Francis Bond

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Outline

- 1 Classifiers
- 2 Aspect
- 3 Serial Verbs
- 4 Subject and Topic
- 5 Sentence Final Particles (SFP)

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What Are Classifiers?

- Classifiers categorize nouns based on salient properties.
- Found in noun phrases (NPs) with modifiers like quantifiers.
- Essential for grammaticality when quantifiers (e.g., numerals) are present.
- Classifiers categorize by physical, social, or functional traits.
- Used with numerals, vague quantifiers like “some”, or even alone.

The Role of Classifiers in NPs

- Classifier slot required for quantifier NPs.
- Used with vague quantifiers like “many” and “some.”
- Classifiers can appear with or without a noun.
- Classifier systems categorize referents based on properties like shape or function.
- In certain NPs, the classifier is mandatory for grammaticality.

Classifier Phrases

- Classifier and quantifier form a grammatical sub-phrase.
- Quantifier and classifier appear adjacent to each other.
- Can sometimes occur alone without a noun.
- These classifier phrases are common across East and Southeast Asian languages.
- Example languages include Thai, Malaysian, and Mandarin.

Examples: Thai and Malaysian

- Thai and Malaysian demonstrate similar classifier behaviors.
- Classifiers are used next to quantifiers.
- Classifier-quantifier pairs can occur without a noun in responses.
- Questions often require classifiers in these languages.

(1) Mǎa kǐi tua?
dog how.many CL:animal
‘How many dogs?’

(2) (Mǎa) sǎam tua.
dog three CL:animal
‘Three (dogs).’

Examples: Malaysian

- Similar behavior with quantifiers and classifiers in Malaysian.
- Classifiers appear adjacent to quantifiers.
- Noun can be omitted in responses.

(3) Berapa ekor anjing?
how.many CL:animal dog
‘How many dogs?’

(4) Tiga ekor (anjing).
three CL:animal (dog)
‘Three (dogs).’

Sortal Classifiers

- **sortal** classifiers are the prototypical numeral classifiers
- they pick out salient features of the noun they classify
 - kinds**, that is, semantic classes
 - shape** (or size)
- **general** classifiers are used as follows:
 - for the residue of nouns
yume-o 1-tu mita “I saw a dream”
 - as a default when no particular feature is salient
hon-o chōdo 1-tu katta “I bought exactly one book”
 - for underspecified referents
nantoka-ga 1-tu hoshii “I want one thingamajig”

(Bond and Paik, 2011)

Measure (Mensural) vs. Sortal Classifier Constructions

- Measure constructions quantify "mass" substances (e.g., water, butter).
- Sortal constructions categorize countable items.
- Measure constructions refer to shape-based or container units.
- Sortal categorize objects based on properties like function or shape.
- Sortal systems apply broadly, while measure constructions are limited.

Examples of Measure Constructions

- Measure constructions often use shape-based units.
- Examples: "two cups of water," "two grains of sand."
- Unit counter constructions group items (e.g., "two pairs of shoes").
- Measure constructions differ from classifiers structurally.

(5) 二 杯 の 水
ni hai no mizu
two CL.cup of water
two cups of water

Unit Counter (Group) Constructions

- Used for collective or group items (e.g., "two pairs of shoes").
 - Unit counters single out items from a larger set.
 - Structurally similar to classifiers but serve different purposes.
 - Examples include: "two sets of furniture," "two bunches of bananas."
 - Not so common in Japanese
 - Very common in English and Thai
- (6) *a flock of birds*
- (7) *a herd of cows*
- (8) *a mob of protestors*

Repeater Constructions

- Nouns can act as their own classifier in repeater constructions.
- Common when no specific classifier exists.
- Often seen in languages like Thai and Lao.
- Repeater constructions fill the classifier slot with the noun itself.

(9) Mii taw dòòk-maj cak taw?
there.is vase flower how.many vase
'How many vases are there?'

Specialized Classifiers

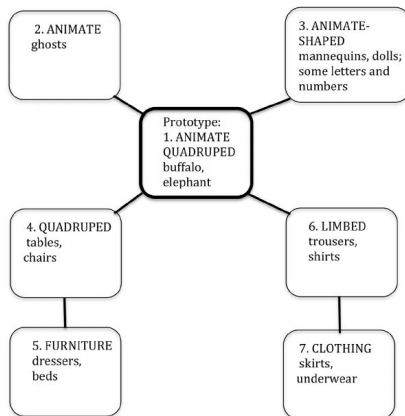
- Some languages, like Korean and Khmer, have specialized classifiers.
- These may be used for royalty, teachers, or monks.
- Classifiers categorize people by social status in some languages.

(10) 何 名 様 ですか
nan mei sama desu ka
what CL.people.HON HON is Q
How many people?

(11) 三 人 です
san nin desu
three CL.people is
Three

Prototype and Extensions in Classifiers

- Classifiers often exhibit polysemy (multiple related meanings).
- A central, or "prototype," meaning extends to other related uses.
- Example: Thai *tua* applies to animals but extends to furniture and clothing.



Classifiers can indirectly count

- (12) 2-tsū-no denshimēru (Japanese)
 2-tong-ui imayl (Korean)
 2-CL.message-of email
 2 pieces of email
- (13) 3-tsū-no tegami (Japanese)
 3-tong-ui phyenci (Korean)
 3-CL.message-of letter
 3 letters
- (14) 3-mai-no tegami
 3-chang-ui phyenci (Korean)
 3-CL.flat-of letter
 {a 3 page letter/*3 letters}

(Bond and Paik, 1997)

Conclusion

- Classifiers categorize nouns based on physical, social, or functional traits.
- Found widely in East and Southeast Asian languages.
- Classifier systems vary from general to highly specialized.
- Classifiers are essential to noun phrase structure in these languages.

Outline

- 1 Classifiers
- 2 Aspect**
- 3 Serial Verbs
- 4 Subject and Topic
- 5 Sentence Final Particles (SFP)

Aspect in East and Southeast Asian Languages

- Many East and Southeast Asian languages lack tense marking.
- Verbs are often marked for aspect instead of tense.
- Aspect marking can occur through suffixes, auxiliary verbs, or particles.
- Aspect focuses on the time structure of an event, not its time reference.
- Aspect can indicate whether an action is ongoing, completed, or habitual.

What is Aspect?

- Aspect refers to the temporal structure of events.
- Different from tense, which indicates time reference (past, present, future).
- Example: English progressive aspect shows ongoing action (e.g. "is talking").
- Many East and Southeast Asian languages mark aspect using various strategies.
- Common aspect categories include perfective, imperfective, progressive, and habitual.

Perfective vs. Imperfective Aspect

- Perfective aspect presents an event in its entirety.
- Imperfective aspect highlights the internal structure of an event.
- Imperfective can indicate ongoing (progressive), repeated (iterative), or habitual actions.
- Example: English progressive *You are talking* is imperfective.
- Perfective focuses on completion, while imperfective focuses on duration or repetition.

Types of Imperfective Aspect

Progressive	Ongoing action	<i>She is running</i>
Iterative	Repeated actions	<i>The light is flashing</i>
Durative	Actions that take time	<i>He was sitting</i>
Habitual	Regularly occurring actions	<i>She runs every morning</i>

Imperfective aspects provide detailed temporal structure.

Aspect in Sinitic Languages: Overview

- Aspect in Mandarin and Cantonese is marked using suffixes and particles.
- Different markers for perfective, progressive, continuous, and experiential aspects.
- Aspect marking is independent of tense (past, present, future).

Aspect Type	Mandarin		Cantonese	
Perfective	V-le	了	V-jo2	咗
Progressive	zai-V	在	V-gan2	緊
Continuous	V-zhe	着	V-jyuh6	住
Experiential	V-guo	过	V-gwo3	過
Delimitative	V-yī-V	一	V-haa5	吓
Habitual			V-hoi1	開

Table: Aspect markers in Mandarin and Cantonese

Perfective Aspect in Mandarin

- Perfective *le* indicates a completed action.
- Used in various contexts: past, future, or conditional events.
- Example: Conditional use of *le* in a future event.

(15) Tā kāi-le mén, nǐ jiù jìn-qu.
3sg open-perf door 2sg then enter-go
(Once) she opens the door, you go in.

Perfective Aspect in Mandarin (continued)

- Perfective *le* can mark completion in different contexts.
- Can occur with specific objects, durations, or numbers of actions.
- Example: Perfective aspect with a stative verb (tall) and quantifier (a little).

(16) Nǐ gǎo-le yīdiǎn.
2sg tall-perf a.little
You've got taller (a little).

Examples of Imperfective Aspects

(17) Wòhng síujé góng-gán dihnwá.

Wong miss talk-prog telephone

Miss Wong is talking on the phone.

(18) Léih hái Hēunggóng jyuh-gwo géi loih a?

you in Hong Kong live-exp how long prt

How long did you live in Hong Kong?

Cantonese: Habitual Aspect

- Habitual aspect *hōi* marks regular or customary actions.
- Example: Habitual actions in present or past tense.

- (19) Kéuih jouh-hōi jūngdím ge.
she work-hab part.time prt
‘She normally works part-time.’
- (20) Kéuih yíhchìhn jā-hōi Bēnsí ge.
she before drive-hab Benz prt
‘She used to drive a Mercedes.’

Delimitative Aspect in Cantonese

- Delimitative *háh* indicates doing something "for a while."
- Often rendered as "have a V" in English.
- Example: "Have a look" or "have a walk."

- (21) Tái-háh.
look-del
Have a look.
- (22) Yāusīk-háh.
rest-del
Have a rest.

Aspect Marking in Malaysian

- Aspect in Malaysian is marked by separate words before the verb.
- *Sudah* (perfective) and *sedang* (progressive) are common markers.
- Perfective *sudah* indicates completion, used for past or future events.
- *Sedang* marks ongoing action, similar to the progressive in other languages.

Examples of Malaysian Aspect

- (23) Saya pun sudah beritahu juga semalam.
I focus perf tell also yesterday
I already told (you) yesterday.
- (24) Bila saya masuk, dia sedang membaca.
when I enter he prog read
When I came in, he was reading.

- This is carried over into Singapore Colloquial English (Singlish)
 - **habitual**: *always*
 - **perfect**: *already*
 - **progressive**: *still*

(25) Eat already? “Have you eaten”

(26) Eat before class “I ate before class”

(27) He always late one. “He is habitually late”

(28) Your phone still spoil ah? “Your phone is still broken?”

Conclusion on Aspect in East and Southeast Asian Languages

- Aspect marking is diverse across languages, with a variety of methods.
- Suffixes, particles, and separate words are used to indicate aspect.
- Common aspect categories include perfective, progressive, habitual, and delimitative.
- Each language has unique strategies for expressing temporal structure.

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Introduction to Serial Verb Constructions

- Serial verb constructions (SVCs) involve two or more verbs in a single clause.
- No conjunctions are used between the verbs.
- All verbs share the same grammatical subject.
- Common in East and Southeast Asian languages.
- Serial verb constructions serve various grammatical functions.

Loose vs. Tight Serialization

- **Loose serialization:** Verbs are loosely connected, depicting sequences of events.
- **Tight serialization:** Verbs are tightly connected, representing a single event.
- In loose serialization, verbs may have independent arguments.
- Tight serialization often involves motion or posture verbs.
- Cultural factors can affect acceptability of tight serialization.

Example of Loose Serialization (Lao)

- Loose serialization in Lao:

(29) Dèèng paj³ talaat⁵ sùu⁴ khùang¹ maa² hùan² lèèw⁴.

Deng go market buy stuff come house perf

Deng went to the market, bought some stuff, (and) came home.

- Verbs describe sequential actions.
- Perfective aspect marker applies to the entire sequence.

Example of Tight Serialization (Thai)

- Tight serialization in Thai:

(30) Dèk wîng pay síi khaŋm.
child run go buy candy

The child ran (and) bought candy.

- Verbs depict a single unitary event.
- Verbs share the same subject and object.

Cultural Factors in Tight Serialization

- Cultural norms influence acceptability of verb combinations.
- Example: Playing musical instruments in Lao.

(31) nang¹ tii³ lanaat⁴.
sit beat lanaat
Play the lanaat sitting.

(32) nòòn² tii³ lanaat⁴.
lie beat lanaat
Play the lanaat lying down.

- Playing the lanaat lying down is culturally unexpected.
- Context can make such constructions acceptable.

Serial Verbs and Directional Verbs (Thai)

- Verbs like ‘go’ and ‘come’ often indicate direction in serial verb constructions.
- These directional verbs function similarly to adverbs.
- Example from Thai:

(33) Khǎw klàp maa lɛ́ɛw.
3sg return hither(come) perf
‘She came back already.’

(34) Khǎw klàp pay lɛ́ɛw.
3sg return away(go) perf
‘She went back already.’

Three-Verb Serialization in Lao

- Serial verb constructions can include three verbs.
- First verb: manner of motion.
- Second verb: absolute direction (e.g., ‘up’ , ‘down’).
- Third verb: direction relative to the speaker (e.g., ‘come’ , ‘go’).

(35) Laaw2 ñaang1 khùn5 maa2.

3sg walk go.up come

‘She walked up here.’

Verb-Prepositions in Serial Constructions

- Certain verbs take on grammatical functions similar to prepositions.
- Verbs like ‘give’ , ‘use’ , ‘go’ mark arguments such as beneficiaries or instruments.
- Example from Lao: haj5 ‘give’ functions as ‘for’ .

(36) Laaw2 nùng1 khaw5 haj5 khòòj5.
3sg steam rice give(=for) 1sg
He steamed some rice for me.

Comitative Use of Serial Verbs (Lao)

- Comitative ‘with’ often expressed using a serial verb construction.
- Example from Lao:

(37) Khan2 haw2 juu1 nam2 mèè1-thaw5.
if we live accompany(=with) mother-old
‘If we live with your mother.’

- Verb ‘accompany’ acts like a preposition meaning ‘with’ .

Serial Verbs in Vietnamese and Hmong

- Verb serialization marks locutionary topics in some languages.
- Example from Vietnamese: ‘arrive’ used for topics.

(38) Chúng tôi nhắc đến anh luôn.
pl 1(excl.) recall arrive(=about) brother often
‘We often speak about you.’

Serial Verbs form a cline in Japanese

- Some serial verbs are completely lexicalized
 - 取り締まる *torishimaru* “manage (lit: take-squeeze)”
 - Some serial verbs are almost aspectual (like English particle verbs)
 - 食べあげる *tabe-ageru* “eat up (lit: eat rise)”
 - 歩きだす *aruki-dasu* “begin to walk/walk out (lit: walk leave)”
 - Some serial verbs are completely syntactic
 - 食べ終わる *tabe-owaru* “finish eating (lit: eat finish)”
 - 歩きはじめる *aruki-hajimeru* “start walking (lit: walk start)”
- ! Arguments can be transferred (in certain cases)
- 大阪を食べ歩く *osaka-wo tabe-aruku* “stroll around Osaka eating (lit: Osaka walk eat)”
 - 寿司を食べ歩く *sushi-wo tabe-aruku* “stroll around eating sushi (lit: sushi walk eat)”

(Hashimoto and Bond, 2005)

Summary of Serial Verb Constructions

- Serial verb constructions involve multiple verbs sharing a subject.
- They are common in East and Southeast Asian languages.
- They can express sequences, direction, beneficiary roles, and more.
- Loose serialization refers to sequences of events.
- Tight serialization depicts a single, unified event.

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What is a Subject?

- NP preceding the verb holds a privileged position.
- English examples:
 - (39) I go
 - (40) He goes.
 - (41) She seems happy.
- Distinction of pronouns: *I* vs. *me*, *they* vs. *them*.
- Subject impacts verb agreement, e.g., -s in present tense.
- Subjects can be agents, patients, experiencers, etc.

Subject in Other Languages

- Subject agreement in European languages.
- Case marking systems indicate subject (nominative).
- Verbs agree with subjects in person and number.
- Passive constructions highlight subjects:
(42) Maxine was elected.
- Traditional view: Subjects are universal in human languages.

Introduction to Topic

- Topics define what the sentence is "about."
- Topics often overlap with subjects in English.

(43) Cats eat mice (topic = cats).

- In many Asian languages, topic and subject diverge.
- Topic prominence is perhaps more important than subject prominence in East Asian languages.

Topic Prominence: Thai

- Flexible word order: OSV, SOV common in Thai.
- Example of OSV:

(44) Pàakphanang khâaw raw kin ngâay.
Pakphanang rice we eat simple
(In) Pakphanang, we ate simply.

- Thai is typically SVO but informal sentences deviate.
- Widespread subject ellipsis in informal contexts.
- Pragmatic ordering of sentences.

Topic Prominence: Mandarin

- Topics always appear sentence-initially.
- Example of topicalized object:

(45) Zhāngsān wǒ yǐjīng jiàn-guo le.
Zhangsan I already see-exp crs
Zhangsan, I've already seen (him).

- Subject ellipsis possible when context is clear.
- Time and locative phrases often function as topics.
- Example: *In Taipei, one can eat well.*

Topic vs. Subject in Mandarin

- Flexible topic-comment structure.
- Double-subject construction in Mandarin.
- Example:

(46) Xiàng bizi cháng.
elephant nose long
'Elephants, (their) noses are long.'

- Topic sets the frame for comment on the subject.
- The topic-comment structure is common and unmarked.

Topic and Subject in Japanese

- *wa* marks topics, *ga* marks subjects.
- Example of *wa/ga* distinction:

(47) hi wa nobor-u.
sun top rise-pres
The sun rises (general)

(48) hi ga nobor-u.
sun top rise-pres
The sun is rising (specific)

- *wa* used in contrastive and generic statements.
- *ga* indicates a focus on the event as a whole.
- Adverbial topics: time/place phrases marked by *wa* (or unmarked)

Tagalog Trigger System

- Tagalog has a flexible trigger system.
- Trigger denotes definiteness and is marked on verbs.
- Example of Actor Trigger (AT):

(49) Mag-aalis ang bata ng laruan sa kahon.
 at-will.take.out trig child pat toy dir box
 ‘The child will take a toy out of the box.’

- Verb affixes indicate the role of the trigger NP.
- The system challenges traditional subject/topic definitions.

Actor vs. Undergoer in Acehnese

- Acehnese verbs take person-marking prefixes or suffixes.
- Prefixes for actors, suffixes for undergoers.
- Example of Actor:

(50) Geu-jak.
3-go
'He/she goes.'

- Example of Undergoer:

(51) Lôn rhët-lôn.
1sg fall-1sg
'I fall.'

- Acehnese highlights actor/undergoer, not subject/object.

Summary: Subject and Topic

- Subject prominence is common in European languages.
- Topic prominence is more evident in East Asian languages.
- Topic and subject do not always align.
- Languages like Tagalog and Acehnese challenge traditional subject concepts.
- Ongoing debates in linguistics about the universality of subjects.

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Illocutionary Particles Overview

- Express speaker's immediate emotions, thoughts, desires.
- Common in Southeast Asian languages.
- Often sentence-final.
- Speech acts: requesting, questioning, persuading.
- Express emotional responses: surprise, doubt, hesitation.

Cantonese Sentence-Final Particles

- Cantonese has over 30 basic particles.
- More than 100 particle combinations.
- Particles add subtle nuances to meaning.
- Function like English intonation but more explicit.
- Rich particle usage typical of Southeast Asian languages.

Examples of Cantonese Particles

Particle		Meaning
呀	ā	lively statement, question
呀	áh	seeking confirmation
噏	ak	abrupt disagreement
啦	lā	requesting, seeking agreement
囉	lo	emphasizing current situation
咩	mē	expressing surprise
㗎	wo	informative, noteworthy
㗎	bo	exclamatory
喇	la	current relevance, advice
㗎	ja	cheeky, intimate

Examples of Cantonese Particles in Questions

- *áh*: surprise or disapproval.
- *há*: presupposing agreement.
- *mē*: surprise, often in rhetorical questions.

(52) Léih gú gam yúhngyih áh?
you guess so easy prt
‘You think it’s that easy, do you?’

Directive Particles

- *la*: emphasizes a point of current relevance.
- *lo*: invites agreement or cooperation.
- Example sentences:

(53) Taaí chóuh la, ngóh fan m̃h dóu.
too noisy prt I sleep not prt
I can't sleep, it's too noisy.

(54) Gám jauh dāk lo.
so then okay prt
'That'll be all right, won't it?'

Particles in Everyday Use

- Particles appear very frequently in Cantonese speech.
- Average occurrence: every 1.5 seconds in conversation.
- Similar usage in Thai, Lao, and Vietnamese.
- Less frequent in formal settings or written texts.
- Frequency decreases in scientific or historical writing.

Mandarin Sentence-Final Particles

- Mandarin has a smaller set of particles than Cantonese.
- Seven common particles in Mandarin.
- Example: *ma* (question), *ba* (suggestion), *ne* (emphasis).
- Particles are typical of informal, conversational settings.
- Estimated frequency: every six seconds in speech.

Examples of Mandarin Particles

- (55) Nǐ xǐhuan chī ma?
you like eat prt
'Do you like to eat?'
- (56) Wǒmen qù ba.
we go prt
'Let's go.'
- (57) Tā ne?
he prt
'What about him?'

These are also common in Singlish I

- (58) You must go lah. “You must go, okay?”
- (59) I already said lor. “I already told you, see?”
- (60) We’re about to go already. Your friend leh¹?
“We’re about to go already. What about your friend” (question)
- (61) Eh, don’t like that leh³. We need you here to play mahjong.
“Don’t be like that. We need you here to play mahjong.”
(persuasive)
- (62) This one easy mah. “This one is easy, you know?”
- (63) You sure can do it meh? “Are you really sure you can do it?”
- (64) Don’t blame me what. “It’s not my fault, you know?”
- (65) She always like that one. “She’s always like that, you know?”

(Chow et al., 2024)

Conclusion: Importance of Particles

- Particles convey subtle shades of meaning in speech.
- Express emotions, intentions, and social cues.
- Different languages utilize particles with varying frequencies.
- Illocutionary particles are an areal feature of Southeast Asia.
- Essential in mastering conversational fluency in these languages.

Conclusions

- SEA languages have many features not found in Indo-European languages
- They package up meaning in different ways

References I

- Francis Bond and Kyonghee Paik. 1997. Classifying correspondence in Japanese and Korean. In *3rd Pacific Association for Computational Linguistics Conference: PACLING-97*, pages 58–67. Meisei University, Tokyo, Japan. URL <https://fcbond.github.io/static/pdf/1997-pacling-cl.pdf>.
- Francis Bond and Kyonghee Paik. 2011. Sortal classifiers in Japanese and Korean. In Aesun Yoon and Eunyoung Lee, editors, *Recent Trends in Language and Knowledge Processing*. Hankukmunwhasa, Seoul.
- Siew Yeng Chow, Chang-Uk Shin, and Francis Bond. 2024. This word mean what: Constructing a Singlish dictionary with ChatGPT. In *Proceedings of the 2nd Workshop on Resources and Technologies for Indigenous, Endangered and Lesser-resourced Languages in Eurasia (EURALI) @ LREC-COLING 2024*, pages 41–50. ELRA and ICCL, Torino, Italia. URL <https://aclanthology.org/2024.eurali-1.7>.

Chikara Hashimoto and Francis Bond. 2005. A computational treatment of V-V compounds in Japanese. In Stefan Müller, editor, *The Proceedings of the 12th International Conference on Head-Driven Phrase Structure Grammar*, pages 143–156. CSLI Publications, Stanford. URL <https://proceedings.hpsg.xyz/issue/view/30>.